CS 6515: Graduate Algorithms

Lecture 9: Linear Programming: Models and Geometry

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Fall 2026

1 Announcements

1. Exam 1 will be graded by next week. We will normalize scores so that median is 86.
2. HW3 will be released soon.

2 Part 2: optimization as an algorithmic tool

In Part 1, we designed algorithms using divide and conquer, dynamic programming, and graph structure. Part 2 adds a different viewpoint: *describe the feasible solutions and the objective mathematically, then exploit the structure of the resulting optimization problem.*

Block	Main questions
Linear programming (9–11)	How do we model a problem with linear constraints? How can we certify that a solution is optimal?
Convex optimization (12–13)	What survives beyond linear functions? How does gradient descent find an approximately optimal solution?
Approximation algorithms (14–16)	When exact discrete optimization is difficult, how can relaxations, rounding, and lower bounds give provably good solutions?

Connections to things we already know. Maximum flow will be our first graph example of an LP. The knapsack FPTAS from Lecture 5 was already a preview of approximation algorithms. Later in Part 2, we will use LPs as relaxations of discrete problems. For convex optimization, the input and the objective are fixed and available throughout the computation.

3 What is a linear program?

A linear program (LP) has real-valued variables, a linear objective, and linear equality or non-strict inequality constraints. Throughout the LP lectures, our *canonical maximization form* is

$$\begin{array}{ll} \text{maximize} & c^\top x \\ \text{subject to} & Ax \le b, \\ & x \ge 0, \end{array} \qquad A \in \mathbb{R}^{m\times n}, \quad b \in \mathbb{R}^m, \quad c \in \mathbb{R}^n.$$

The inequalities are coordinatewise. The j-th row is $\sum_{i=1}^{n} A_{ji}x_i \le b_j$. For computational statements, the input coefficients are rational.

Definition (feasible region). The feasible region is the set of vectors satisfying *all* constraints:

$$P = \{x \in \mathbb{R}^n : Ax \le b,\ x \ge 0\}.$$

An optimal solution is a feasible vector with the best objective value.

3.1 Running example: making smoothies

We have 3 units of oranges, 10 of bananas, and 6 of strawberries. We can make fractional liters; the ingredients and revenue per liter are:

Smoothie	Oranges	Bananas	Strawberries	Revenue
1	1	2	0	\$5
2	0	1	2	\$4
3	1	1	1	\$6

Let x_i be the number of liters of smoothie i. The LP is

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 6x_3 \\ \text{subject to} \quad & x_1 + x_3 \le 3 \qquad \text{(oranges)}, \\ & 2x_1 + x_2 + x_3 \le 10 \qquad \text{(bananas)}, \\ & 2x_2 + x_3 \le 6 \qquad \text{(strawberries)}, \\ & x_1, x_2, x_3 \ge 0. \end{aligned} \tag{1}$$

The vector $(3, 3, 0)$ is feasible and earns 27. This proves OPT ≥ 27, but *does not yet prove optimality.*

Exercise. Write the matrix A and vectors b, c. What changes if smoothies can only be sold in whole-liter batches?

4 Modeling familiar problems and changing forms

4.1 Maximum flow is an LP

For a directed network, give each edge a variable f_e. As in Lecture 8, the value is net flow out of the source:

$$\begin{aligned} \text{maximize} \quad & \sum_{(s,v)\in E} f_{sv} - \sum_{(v,s)\in E} f_{vs} \\ \text{subject to} \quad & \sum_{(u,v)\in E} f_{uv} = \sum_{(v,w)\in E} f_{vw} \quad (v \notin \{s,t\}), \\ & 0 \le f_e \le c_e \quad (e \in E). \end{aligned}$$

There are $m = |E|$ variables and $\mathcal{O}(m+n)$ constraints. An LP formulation does not replace the specialized flow algorithms; it places flow in a broader optimization framework.

Real variables versus integer variables. The LP allows fractional flows. Lecture 8 proved that integer capacities guarantee an *integral optimal flow.* This is special structure: an arbitrary LP need not have an integral optimum.

4.2 Equivalent LP forms

Feature	Conversion to canonical form
Minimize $c^\top x$	Maximize $-c^\top x$, then negate the value.
$a^\top x \geq d$	Write $-a^\top x \leq -d$.
$a^\top x = d$	Write both $a^\top x \leq d$ and $-a^\top x \leq -d$.
Unrestricted variable z	Substitute $z = z^+ - z^-$, with $z^+, z^- \geq 0$.

These transformations increase the formulation size only polynomially. Another useful form is $Ax + s = b,\ x, s \geq 0$, obtained by adding *slack variables* $s = b - Ax$.

What is not a plain LP? Constraints such as $x_1 x_2 \leq 1$, an objective such as $x_1^2 + x_2^2$, or requirements $x_i \in \{0, 1\}$ are not part of an LP. We also do not allow strict inequalities: maximizing x subject to $x < 1$ has supremum 1, but no maximizing point.

Exercise. Convert $\min(2u - v)$, subject to $u - v = 3$ and $u \geq 0$, into canonical form when v is unrestricted.

5 The geometry of an LP

Definition (convex set). A set $P \subseteq \mathbb{R}^n$ is convex if, for every $x, y \in P$ and $\lambda \in [0, 1]$, the point $\lambda x + (1 - \lambda)y$ also belongs to P. Equivalently, the whole line segment between any two points of P stays in P.

Theorem (LP feasible regions are convex). The feasible region of every LP is convex.

Proof. If $Ax \leq b$ and $Ay \leq b$, then

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay \leq \lambda b + (1 - \lambda)b = b.$$

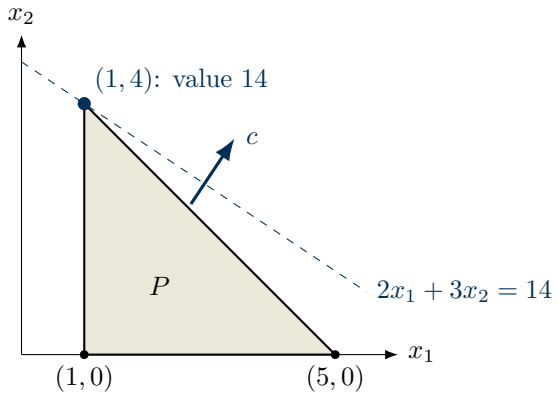
Nonnegativity is also preserved. Equalities are preserved by the same calculation.

5.1 A two-dimensional picture

Consider

$$\max\ 2x_1 + 3x_2 \quad \text{subject to} \quad x_1 + x_2 \leq 5, \quad x_1 \geq 1, \quad x_2 \geq 0.$$

Its feasible region is the triangle below. Lines of constant objective have equation $2x_1 + 3x_2 = q$; increasing q moves them in direction $c = (2, 3)$.



The segment stays feasible. The best objective line touches a vertex.

If the objective were $x_1 + x_2$, an entire edge would be optimal. Thus an optimum need not be unique.

6 Vertices and the three possible outcomes

An LP can be:

- **infeasible:** no feasible point, e.g., $x \geq 2,\ x \leq 1$;
- **unbounded in objective:** feasible points have arbitrarily large objective, e.g., maximize x subject to $x \geq 0$;
- **feasible with finite optimum:** then an LP attains its optimum.

An unbounded *feasible region* does not imply an unbounded objective: maximize $-x$ subject to $x \geq 0$.

Definition (tight constraint and vertex). An inequality $a^\top x \leq d$ is tight at x if $a^\top x = d$. A feasible point of a polyhedron in $\mathbb{R}^n$ is a vertex (or basic feasible solution) if the normals of its tight constraints span $\mathbb{R}^n$: there are n linearly independent tight constraints.

Nonnegativity constraints count: write $x_i \geq 0$ as $-x_i \leq 0$. A vertex can have more than n tight constraints; independence matters.

Theorem (a basic optimal solution). If an LP in canonical form $Ax \leq b,\ x \geq 0$ is feasible with finite optimal value, it has an optimal solution at a vertex.

Geometric intuition. If fewer than n independent constraints are tight, there is a nonzero direction parallel to all of them. A sufficiently small move in either direction remains feasible. At an optimum, this direction cannot improve the objective, so the objective is constant along it. Move until another independent constraint becomes tight, and repeat. Nonnegativity prevents a whole line from remaining feasible in both directions forever.

A useful caveat. Without the canonical-form assumption, a polyhedron can have no vertices. For example, the half-plane $\{(u, v) : v \geq 0\}$ has none, although $\max(-v)$ has finite optimum 0. Converting to canonical form restores the vertex statement in the new variables.

Exercise. For the two-dimensional triangle, identify two independent tight constraints at each vertex. At $(3, 3, 0)$ in the smoothie LP, identify three independent tight constraints.

7 Why LPs are algorithmically useful

Theorem (polynomial-time solvability, used as a black box). For rational input, an LP can be solved in time polynomial in its total binary encoding length: report infeasibility or unboundedness, or return an exact rational optimum when the optimal value is finite.

The ellipsoid method and interior-point methods underlie polynomial-time LP algorithms. We will use LP solving as a tool, not develop these algorithms.

7.1 Why can the answer be written with polynomially many bits?

At a vertex, select n independent tight constraints, including nonnegativity constraints if needed. They give a nonsingular system

$$Bx = d.$$

By Cramer's rule, $x_i = \det(B_i)/\det(B)$, where B_i is obtained by replacing column i of B with d. After clearing rational denominators, determinants of integer matrices have polynomial encoding length: each is a sum of $n!$ products of n entries. Hence a basic optimal solution has polynomial bit length.

This is not an enumeration algorithm. There may be exponentially many choices of tight constraints. The existence of a short answer does not imply that testing every candidate vertex is efficient.

Connection to our knapsack discussion. Polynomial in *input bit length* differs from polynomial in the numerical magnitudes. If a capacity is $B = 2^r$, its encoding uses $r + 1$ bits, whereas a DP taking $\mathcal{O}(nB)$ time is pseudo-polynomial. An LP algorithm's polynomial guarantee is in the encoding length; do not read this theorem as saying the running time depends only on the numbers of variables and constraints.

Exercise. An LP in n variables has box constraints $0 \le x_i \le 1$ and only k other linear inequalities. Show that any vertex has at most k coordinates strictly between 0 and 1.

8 How do we prove a proposed solution is optimal?

Return to the smoothie LP.

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 6x_3 \\ \text{subject to} \quad & x_1 + x_3 \le 3 \qquad \text{(oranges)}, \\ & 2x_1 + x_2 + x_3 \le 10 \qquad \text{(bananas)}, \\ & 2x_2 + x_3 \le 6 \qquad \text{(strawberries)}, \\ & x_1, x_2, x_3 \ge 0. \end{aligned} \tag{2}$$

The feasible solution

$$x = (3, 3, 0) \qquad \text{has value} \qquad 5 \cdot 3 + 4 \cdot 3 = 27.$$

To prove it is optimal, we need an upper bound of 27 for *every* feasible solution.

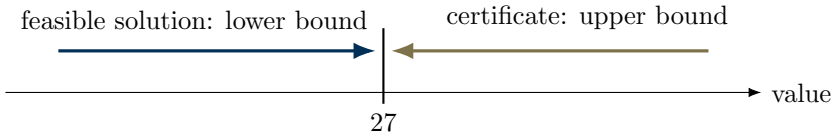
Take five times the orange constraint and two times the strawberry constraint:

$$5(x_1 + x_3) + 2(2x_2 + x_3) \leq 5 \cdot 3 + 2 \cdot 6 = 27.$$

The left side is $5x_1 + 4x_2 + 7x_3$. Since $x_3 \geq 0$,

$$\boxed{5x_1 + 4x_2 + 6x_3 \leq 5x_1 + 4x_2 + 7x_3 \leq 27.}$$

Equivalently, include one copy of the constraint $-x_3 \leq 0$ to make the left side exactly the objective.



The next question. We found nonnegative weights $(5, 0, 2)$ on the three resource constraints. Can we systematically search for the *best* such upper bound? That search is itself an LP: the **dual**.

Takeaways. An LP separates the model from the algorithm. Convex geometry explains its structure, and a short certificate can prove optimality without requiring us to rerun the optimization algorithm.

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Lecture 10: LP Duality and Max-Flow/Min-Cut

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Fall 2026

Announcements. HW3 is out; due Oct 13th.

Today's plan. We begin with the optimality-certificate discussion left over from Lecture 9, then derive LP duality and apply it to max-flow/min-cut.

1 How do we prove a proposed solution is optimal?

Recall the smoothie LP from Lecture 9:

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 6x_3 \\ \text{subject to} \quad & x_1 + x_3 \le 3 \qquad \text{(oranges)}, \\ & 2x_1 + x_2 + x_3 \le 10 \qquad \text{(bananas)}, \\ & 2x_2 + x_3 \le 6 \qquad \text{(strawberries)}, \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

The solution $x = (3, 3, 0)$ uses 3 oranges, 9 bananas, and 6 strawberries, so it is feasible and has value $5 \cdot 3 + 4 \cdot 3 = 27$. This gives a *lower bound* on the maximum. To prove optimality, we need an upper bound of 27 for *every* feasible solution.

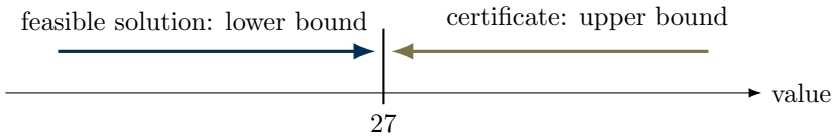
Build an upper bound from the constraints. Take five times the orange constraint and two times the strawberry constraint:

$$5(x_1 + x_3) + 2(2x_2 + x_3) \le 5 \cdot 3 + 2 \cdot 6 = 27.$$

The left side is $5x_1 + 4x_2 + 7x_3$. Since $x_3 \ge 0$,

$$\boxed{5x_1 + 4x_2 + 6x_3 \le 5x_1 + 4x_2 + 7x_3 \le 27.}$$

Equivalently, include one copy of $-x_3 \le 0$ to make the left side exactly the objective. The matching lower and upper bounds prove that $x = (3, 3, 0)$ is optimal.



From one certificate to a general method. The weights $(5, 0, 2)$ give a short *certificate* of optimality. How could we systematically find the best such upper bound? We will treat the nonnegative constraint weights as variables: the search for a certificate is itself an LP, called the **dual**.

2 Deriving the dual: optimize over upper bounds

The smoothie argument used only two facts: nonnegative weighted sums preserve inequalities, and the weighted left side dominates the objective coefficient by coefficient. Let us generalize it to the canonical primal LP

$$(P) \qquad \max\ c^\top x \quad \text{subject to} \quad Ax \le b, \quad x \ge 0.$$

Put a nonnegative weight y_j on row j and add the inequalities:

$$\sum_{j=1}^{m} y_j \left(\sum_{i=1}^{n} A_{ji} x_i \right) \le \sum_{j=1}^{m} y_j b_j.$$

If $\sum_j A_{ji} y_j \ge c_i$ for every variable i, then nonnegativity of x_i gives

$$\boxed{c^\top x \ \le\ x^\top A^\top y \ =\ y^\top A x \ \le\ b^\top y.}$$

Thus every such y certifies an upper bound $b^\top y$. To find the best certificate, solve

$$(D) \qquad \min\ b^\top y \quad \text{subject to} \quad A^\top y \ge c, \quad y \ge 0.$$

Primal maximization LP	Dual minimization LP
n variables x_i	n constraints $(A^\top y)_i \ge c_i$
m constraints $(Ax)_j \le b_j$	m variables y_j
Objective coefficients c_i	Right-hand sides c_i
Right-hand sides b_j	Objective coefficients b_j

Why the signs matter. We need $y \ge 0$ when multiplying the primal inequalities, and $x \ge 0$ when comparing $c^\top x$ with $(A^\top y)^\top x$. Neither sign condition is cosmetic.

3 Weak duality and a worked certificate

Theorem (weak LP duality). For every primal-feasible x and dual-feasible y,

$$c^\top x \le b^\top y.$$

Consequently, if $c^\top x = b^\top y$, then both solutions are optimal.

Proof. The inequality is the chain in the preceding section. Equality leaves no room for either LP to improve.

3.1 Revisiting the smoothie certificate

The three resource weights from our opening example now become variables y_O, y_B, y_S. Applying the general construction gives

$$\begin{array}{ll} \text{minimize} & 3y_O + 10y_B + 6y_S \\ \text{subject to} & y_O + 2y_B \ge 5, \\ & y_B + 2y_S \ge 4, \\ & y_O + y_B + y_S \ge 6, \\ & y_O, y_B, y_S \ge 0. \end{array}$$

The first constraint corresponds to smoothie 1, the second to smoothie 2, and the third to smoothie 3.

Example. Our opening weights $y = (5, 0, 2)$ satisfy all three dual constraints: $5 \ge 5$, $4 \ge 4$, and $7 \ge 6$. Their dual value is $3 \cdot 5 + 10 \cdot 0 + 6 \cdot 2 = 27$, matching the primal value at $x = (3, 3, 0)$. The earlier certificate is exactly an application of weak duality: both solutions are optimal.

Economic interpretation. Imagine purchasing all available ingredients at prices y_O, y_B, y_S. Every dual constraint says that the ingredient cost of one liter of a smoothie is at least its revenue. Therefore no production plan can earn more than the total priced inventory. The dual minimizes this valid upper bound.

At the displayed optimum, one banana is unused and the banana price is zero. Smoothie 3 has priced ingredients worth 7, exceeding revenue 6, and is not produced. These observations will become *complementary slackness* in Lecture 11.

Exercise. Check that $(y_O, y_B, y_S) = (3, 1, 2)$ is dual-feasible. What upper bound does it give? Why is a feasible certificate not necessarily the best certificate?

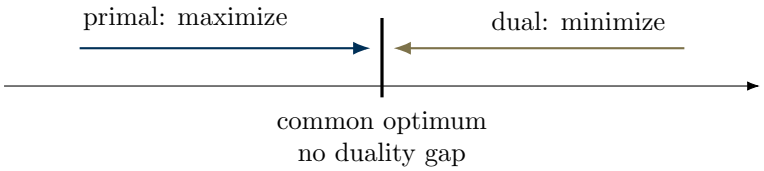
4 Strong duality and other LP forms

Theorem (strong LP duality). If the primal LP is feasible with a finite optimal value, then the dual also has an optimum, and

$$\boxed{\max\{c^\top x : Ax \le b,\ x \ge 0\} = \min\{b^\top y : A^\top y \ge c,\ y \ge 0\}.}$$

The symmetric statement holds starting from a finite dual optimum.

Weak duality says every certificate is an upper bound. Strong duality says an *exact* certificate exists. We will prove this in Lecture 11.



Infeasibility is not the same as unboundedness.

- If the primal is unbounded above, the dual is infeasible.
- If the dual is unbounded below, the primal is infeasible.
- The two LPs can *both* be infeasible. Thus the converse of either preceding statement is false.

For example, the primal

$$\max(2x_1 - x_2): \quad x_1 - x_2 \le 1, \quad -x_1 + x_2 \le -2, \quad x \ge 0$$

is infeasible. Its dual requires $y_1 - y_2 \ge 2$ and $-y_1 + y_2 \ge -1$, which is also impossible.

4.1 A compact rulebook

For a *maximization* primal and its minimization dual:

Primal feature	Corresponding dual feature
Constraint $a_j^\top x \le b_j$	$y_j \ge 0$
Constraint $a_j^\top x \ge b_j$	$y_j \le 0$
Constraint $a_j^\top x = b_j$	y_j unrestricted
Variable $x_i \ge 0$	$(A^\top y)_i \ge c_i$
Variable $x_i \le 0$	$(A^\top y)_i \le c_i$
Variable x_i unrestricted	$(A^\top y)_i = c_i$

The objective is still $b^\top y$. Derive these rules by preserving the weak-duality inequality, or first convert to canonical form. The dual of the dual is the primal.

5 Application: max-flow/min-cut from LP duality

Lecture 8 used residual reachability to produce a cut with the same value as a maximum flow. We now obtain the same equality by a different route.

5.1 A path-based flow LP

Let $\mathcal{P}$ be the finite set of all simple directed s-t paths. Give path p a variable x_p, the amount sent along it:

$$\begin{aligned} (P_{\text{flow}}) \quad \text{maximize} \quad & \sum_{p \in \mathcal{P}} x_p \\ \text{subject to} \quad & \sum_{p \ni e} x_p \le c_e \quad (e \in E), \\ & x_p \ge 0 \quad (p \in \mathcal{P}). \end{aligned}$$

Adding the path flows gives an ordinary feasible flow of the same value. Conversely, any feasible flow of nonnegative value can be decomposed into s-t path flows and directed cycle flows; discard the cycles without changing its value. One way to see the decomposition is to add an edge $t \to s$ carrying the flow value and decompose the resulting circulation into cycles.

Size warning. There can be exponentially many simple paths. This formulation is for understanding duality, not an instruction to enumerate all paths. The edge-based LP in Lecture 9 has polynomial size.

5.2 The dual assigns edge lengths

There is a dual variable y_e for each capacity constraint:

$$\begin{aligned} (D_{\text{length}}) \quad \text{minimize} \quad & \sum_{e \in E} c_e y_e \\ \text{subject to} \quad & \sum_{e \in p} y_e \ge 1 \quad (p \in \mathcal{P}), \\ & y_e \ge 0 \quad (e \in E). \end{aligned}$$

Interpret y_e as a length: every s-t path must have length at least 1. Strong duality equates this fractional length optimum with maximum flow.

Cuts give feasible dual solutions. For an s-t cut S, set $y_e = 1$ on edges leaving S and 0 elsewhere. Every s-t path leaves S at least once, and the dual value is exactly the capacity $c(\delta^+(S))$, where $\delta^+(S) = \{(u, v) \in E : u \in S,\ v \notin S\}$.

The missing step is important: *why can a fractional length solution be replaced by a cut without increasing its cost?*

6 From fractional lengths to an actual cut

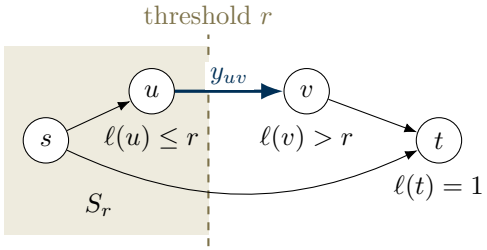
Take any feasible dual lengths $y \geq 0$. Let $d(v)$ be shortest-path distance from s in these lengths, with $d(v) = +\infty$ if unreachable. Define clipped distances

$$\ell(v) = \min\{d(v), 1\}, \qquad \ell(s) = 0, \qquad \ell(t) = 1.$$

The last equality holds because every s-t path has length at least 1. For each threshold $r \in (0, 1)$, define

$$S_r = \{v : \ell(v) \leq r\}.$$

Every S_r is an s-t cut.



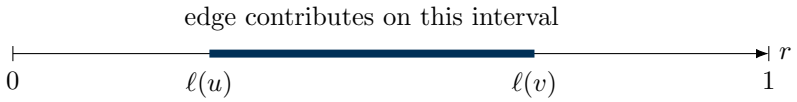
Only directed edges leaving S_r contribute to its capacity.

6.1 For how many thresholds is one edge cut?

An edge $e = (u, v)$ leaves S_r exactly when $\ell(u) \leq r < \ell(v)$. The length of this interval is

$$\max\{0, \ell(v) - \ell(u)\} \leq y_{uv}.$$

Indeed, when $d(u) < 1$, the shortest-path inequality gives $d(v) \leq d(u) + y_{uv}$, and clipping cannot increase the positive difference. When $d(u) \geq 1$ or u is unreachable, $\ell(u) = 1$ and the positive difference is zero.



6.2 Average the cut capacities

Sweep r from 0 to 1. The average cut capacity is at most

$$\begin{aligned}\int_0^1 c(\delta^+(S_r))\,dr &= \sum_{(u,v)\in E} c_{uv}\max\{0, \ell(v)-\ell(u)\} \\ &\le \sum_{e\in E} c_e y_e.\end{aligned}$$

This integral is just the area of a step function: the cut changes only when r passes one of the finitely many vertex distances. Equivalently, take a weighted average of the finitely many cut values, with weights equal to the lengths of the threshold intervals.

At least one cut is no more expensive than the average. Therefore

$$\boxed{\text{some } S_r \text{ satisfies } c(\delta^+(S_r)) \le \sum_e c_e y_e.}$$

The cut can be found deterministically by checking the distinct threshold cuts after computing shortest-path distances.

6.3 Complete the proof

Apply the rounding argument to an optimal dual solution:

$$\begin{aligned}\text{maximum-flow value} &\le \text{minimum-cut capacity} && \text{(the cut bound from Lecture 8)},\\ \text{minimum-cut capacity} &\le \text{optimal length-dual value} && \text{(threshold rounding)},\\ \text{optimal length-dual value} &= \text{maximum-flow value} && \text{(strong LP duality)}.\end{aligned}$$

All inequalities must be equalities.

What is integral here? The length dual has an optimal 0/1 solution corresponding to a cut, even for nonnegative real capacities. This is different from the integrality of the *flow*, which in Lecture 8 required integral capacities.

Preview of approximation algorithms. Here rounding loses nothing. For other discrete problems, an LP may have a better fractional value than any integral solution. We will aim to round while losing only a bounded multiplicative factor.

Exercise. Given nonnegative lengths y_e, explain how one Dijkstra run can either verify every path constraint in D_{length}, or exhibit a violated one. Why do we not need to list all paths?

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Lecture 11: Why Strong Duality Holds

Instructors: Sahil Singla and Gerandy Brito

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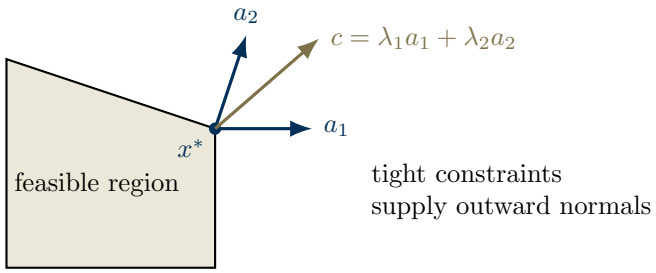
Today's question. Weak duality is an easy calculation. Why must an exact upper-bound certificate exist whenever the primal has a finite optimum? Our proof combines a geometric certificate of infeasibility (Farkas' lemma) with the active-constraint viewpoint.

1 The proof plan

Recall

$$\begin{array}{lllll} (P) & \max c^\top x & Ax \le b, & x \ge 0, \\ (D) & \min b^\top y & A^\top y \ge c, & y \ge 0. \end{array}$$

Let x^* be an optimal primal solution. A constraint $a^\top x \le d$ that is tight at x^* forbids moving in a direction v with $a^\top v > 0$. An inactive constraint has positive slack, so it does not prevent a sufficiently small move.



At an optimum, the objective direction is a nonnegative combination of the active outward normals.

We will justify the picture in three steps:

1. Either c is a nonnegative combination of the active normals, or a separating direction points toward a better feasible solution.
2. A better feasible solution contradicts optimality.
3. The nonnegative coefficients form a dual solution with the same objective value as x^*.

Important bookkeeping. The active normals include those of $-x_i \le 0$. For instance, $\max(-x)$ subject to $0 \le x \le 1$ is optimized at $x = 0$; the relevant tight normal is -1, not the inactive upper bound's normal $+1$.

2 Farkas' lemma: certificates of infeasibility

Given columns $a_1, \ldots, a_k \in \mathbb{R}^d$, their nonnegative combinations form a cone

$$C = \left\{ \sum_{j=1}^{k} \lambda_j a_j : \lambda_j \geq 0 \right\}.$$

The question $M\lambda = p,\ \lambda \geq 0$, with $M = [a_1\ \cdots\ a_k]$, asks whether p belongs to this cone.

Theorem (Farkas' lemma). For any real matrix M and vector p, exactly one of the following systems has a solution:

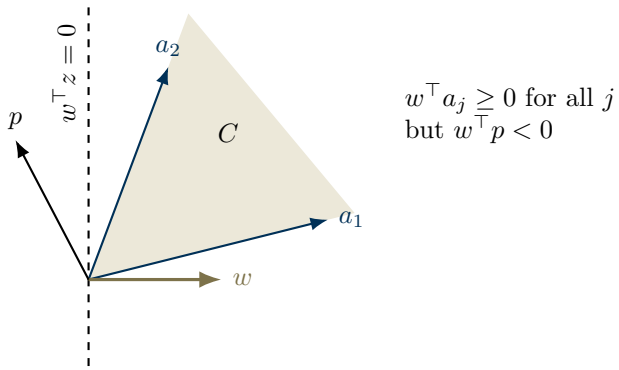
$$\begin{aligned} &\text{(I)} \quad M\lambda = p, \quad \lambda \geq 0; \\ &\text{(II)} \quad M^\top w \geq 0, \quad p^\top w < 0. \end{aligned}$$

The vector w in (II) is unrestricted in sign.

Why they cannot both hold. If they did, then

$$p^\top w = (M\lambda)^\top w = \lambda^\top M^\top w \geq 0,$$

contradicting $p^\top w < 0$.



Example. Take $a_1 = (1, 0)$, $a_2 = (1, 1)$, and $p = (0, 1)$. No nonnegative combination equals p. The certificate $w = (1, -1)$ satisfies $a_1^\top w = 1$, $a_2^\top w = 0$, and $p^\top w = -1$.

The only nontrivial part of Farkas' lemma is that a separating certificate *always* exists when $p \notin C$.

3 The geometric reason: separation

Theorem (separating a point from a closed convex set). Let $C \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let $p \notin C$. There are a vector v and a point $q \in C$ such that

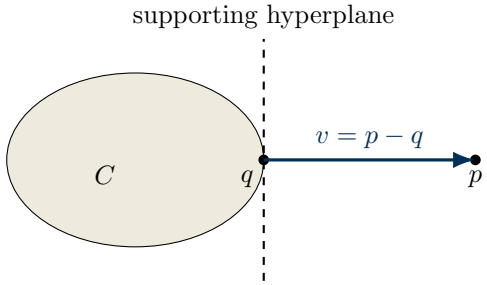
$$v^\top p > v^\top q \geq v^\top z \qquad \text{for every } z \in C.$$

Proof. Choose a closest point $q \in C$ to p, and set $v = p - q$. A closest point exists: after fixing any candidate in C, we can restrict to its closed distance ball around p, which is compact.

For every $z \in C$, convexity keeps $q + \theta(z - q)$ feasible for $0 \leq \theta \leq 1$. If $v^\top(z - q) > 0$, then for sufficiently small $\theta > 0$,

$$\begin{aligned} \|p - (q + \theta(z - q))\|_2^2 &= \|p - q\|_2^2 - 2\theta v^\top(z - q) + \theta^2\|z - q\|_2^2 \\ &< \|p - q\|_2^2, \end{aligned}$$

a contradiction. Thus $v^\top z \leq v^\top q$. Finally $v^\top p - v^\top q = \|p - q\|_2^2 > 0$.



3.1 Finish the proof of Farkas' lemma

We use one standard geometric fact: a cone generated by finitely many vectors is closed (as well as convex). If $p \notin C$, apply separation. Since $0 \in C$, the separating upper bound is nonnegative. Since C contains every nonnegative multiple of each a_j, a finite upper bound requires $v^\top a_j \leq 0$ for every j. Thus the largest value of $v^\top z$ on C is 0, and $v^\top p > 0$. Set $w = -v$: then $M^\top w \geq 0$ and $p^\top w < 0$, as required.

What to remember. Infeasibility of a nonnegative linear combination has a short *linear* witness. The geometric proof explains why the witness exists.

4 Strong duality: the active-constraint argument

Write *all* primal inequalities as

$$a_r^\top x \le d_r \qquad (r = 1, \dots, m+n),$$

where the last n rows are $-x_i \le 0$. Let S be the rows tight at an optimal primal solution x^*. We claim

$$c = \sum_{r \in S} \lambda_r a_r \qquad \text{for some } \lambda_r \ge 0. \tag{1}$$

4.1 If the claim fails, Farkas gives an improving direction

Apply Farkas' lemma to the matrix whose columns are a_r, $r \in S$, with target c. If (1) is impossible, there is w with $a_r^\top w \ge 0$ for all $r \in S$ and $c^\top w < 0$. Let $v = -w$. Then

$$a_r^\top v \le 0 \quad (r \in S), \qquad c^\top v > 0.$$

Consider $x^* + \varepsilon v$:

- Tight constraints stay feasible because $a_r^\top (x^* + \varepsilon v) \le d_r$.
- An inactive constraint has slack $\delta_r = d_r - a_r^\top x^* > 0$. If $a_r^\top v > 0$, choose $\varepsilon \le \delta_r / (a_r^\top v)$; otherwise it imposes no restriction.

There are finitely many rows, so a sufficiently small $\varepsilon > 0$ satisfies all of them, including the nonnegativity rows. But its objective increases by $\varepsilon c^\top v > 0$, a contradiction. This proves (1).

4.2 Turn the coefficients into a dual solution

Give inactive rows coefficient zero. Let $y_j \ge 0$ be the coefficients of the original m rows, and $z_i \ge 0$ the coefficients of $-x_i \le 0$. Then

$$c = A^\top y - z \quad \Longrightarrow \quad A^\top y \ge c.$$

So y is dual-feasible. Every row with positive coefficient is tight; the bound rows have right-hand side 0. Therefore

$$c^\top x^* = \sum_r \lambda_r a_r^\top x^* = \sum_r \lambda_r d_r = b^\top y.$$

Weak duality now proves that y is dual-optimal and the two optimal values coincide. Notice that x^* did not need to be a vertex.

5 What these three LP lectures give us

1. **Modeling.** Real variables and linear constraints express resource allocation and graph problems.
2. **Algorithms.** Polynomial-size rational LPs can be solved in polynomial time in their encoding length.
3. **Certificates.** A feasible solution and a matching-value dual solution prove optimality.
4. **Geometry.** At an optimum, tight constraints block every improving direction. Farkas' lemma converts this fact into multipliers.
5. **Rounding.** A fractional LP optimum can guide the search for an integral solution; max-flow/min-cut is an exact example.

5.1 Bridge to convex optimization

For an LP, the objective has a fixed direction c. For a differentiable nonlinear objective f, the gradient $\nabla f(x)$ gives a local linear approximation:

$$f(x+h) \approx f(x) + \nabla f(x)^\top h.$$

Convexity will make that local information globally useful.

We will next *minimize* a convex function over a convex set. A maximization LP fits this viewpoint by minimizing the convex linear function $-c^\top x$. We will then move beyond linear objectives to examples such as least squares.

5.2 Checks for understanding

Exercise. For $\max(-x)$ subject to $x \le 1,\ x \ge 0$, write its dual and find an optimal primal-dual pair. Which nonnegativity normal supplies the objective direction in the strong-duality proof?

Exercise. Find a feasible LP with a unique optimum for which a tight constraint has zero dual multiplier. Why does this not contradict complementary slackness?

Exercise. Using only weak duality, show that a feasible pair with $b^\top y - c^\top x \le \varepsilon$ certifies $\mathrm{OPT} - c^\top x \le \varepsilon$.

CS 6515: Graduate Algorithms

Lecture 12: Convexity and Gradient Descent

Instructors: Sahil Singla and Gerandy Brito

Fall 2026

From linear to nonlinear optimization. An LP optimizes a linear function over a convex feasible region. We now allow a nonlinear objective, while retaining the geometric property that will make the problem tractable: convexity.

1 The offline optimization problem

We are given one fixed function $f : \mathbb{R}^d \to \mathbb{R}$ and a nonempty convex set $K \subseteq \mathbb{R}^d$. Our goal is

$$\min_{x \in K} f(x).$$

Whenever we write x^*, we assume a minimizer exists. An ε-optimal feasible answer satisfies

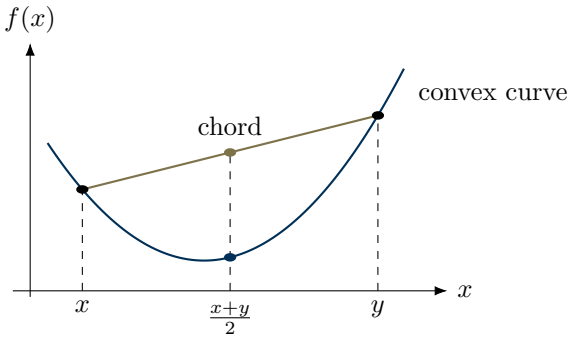
$$f(x) - f(x^*) \leq \varepsilon.$$

The input and objective are fixed throughout the computation. A value/gradient oracle lets us evaluate $f(x)$ and $\nabla f(x)$ at points we choose.

Definition (convex function). A function f is convex on a convex set K if, for all $x, y \in K$ and $\lambda \in [0, 1]$,

$$\boxed{f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y).}$$

The graph lies below its chords.



At the midpoint, $f((x+y)/2) \leq (f(x) + f(y))/2$.

A linear function is both convex and concave. Thus maximizing $c^\top x$ over an LP region is equivalent to minimizing the convex function $-c^\top x$.

2 The first-order view: tangent planes are lower bounds

For vectors, write

$$\langle u, v\rangle = u^\top v = \sum_i u_i v_i, \qquad \|u\|_2 = \sqrt{\sum_i u_i^2}, \qquad \nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_d}\right).$$

We use the Euclidean norm throughout these two lectures.

Theorem (first-order characterization). A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is convex if and only if, for all $x, y \in \mathbb{R}^d$,

$$\boxed{f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle.}$$

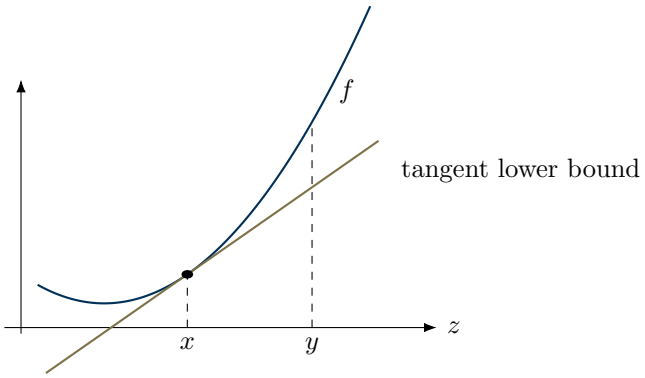
Proof of the direction we will use. For $0 < \theta \leq 1$, convexity gives

$$f(x + \theta(y - x)) \leq (1 - \theta) f(x) + \theta f(y).$$

Rearrange and let $\theta \downarrow 0$:

$$f(y) - f(x) \geq \lim_{\theta \downarrow 0} \frac{f(x + \theta(y - x)) - f(x)}{\theta} = \langle \nabla f(x), y - x\rangle.$$

For the converse, apply the first-order inequality at $z = \lambda x + (1 - \lambda)y$, once to x and once to y. Multiply by λ and $1 - \lambda$ and add: the gradient terms cancel.



The inequality to remember. Putting $y = x^*$ and rearranging,

$$\boxed{f(x) - f(x^*) \leq \langle \nabla f(x), x - x^*\rangle.}$$

This converts objective error into an inner product that we can analyze using the algorithm's update.

3 Examples and why local minima suffice

3.1 A calculus test, when second derivatives exist

For a twice continuously differentiable $f : \mathbb{R}^d \to \mathbb{R}$, convexity is equivalent to

$$\nabla^2 f(x) \succeq 0 \quad \text{for every } x.$$

Here $\nabla^2 f$ is the Hessian matrix, and positive semidefinite means $v^\top \nabla^2 f(x) v \geq 0$ for every vector v. In one dimension this is simply $f''(x) \geq 0$. Restricting f to each line gives the multivariable version of this test.

Example. For least squares,

$$f(x) = \tfrac{1}{2}\|Ax - b\|_2^2, \qquad \nabla f(x) = A^\top (Ax - b), \qquad \nabla^2 f(x) = A^\top A.$$

Since $v^\top A^\top A v = \|Av\|_2^2 \geq 0$, least squares is convex. The transpose $A^\top$ in the gradient is essential.

Example. The function $f(x) = |x|$ is convex but not differentiable at 0. A maximum of affine functions, $f(x) = \max_j (a_j^\top x + b_j)$, is also convex. Thus differentiability is an additional assumption, not part of the definition of convexity.

Theorem (local minimum implies global minimum). If f and K are convex, every local minimum of f on K is a global minimum.

Proof. Suppose $x \in K$ is a local minimum but some $y \in K$ has $f(y) < f(x)$. For every $0 < \theta \leq 1$, the feasible point $z_\theta = (1 - \theta)x + \theta y$ satisfies

$$f(z_\theta) \leq (1 - \theta) f(x) + \theta f(y) < f(x).$$

As $\theta \downarrow 0$, these better feasible points approach x, contradicting local optimality.

Stationary points and constraints. For differentiable unconstrained convex minimization, $\nabla f(x) = 0$ certifies a global minimum by the first-order inequality. The converse also holds for an unconstrained minimum. At a constrained boundary optimum the gradient need not vanish: $\min_{x \in [0,1]} x$ is optimized at 0, with derivative 1.

4 Gradient descent

First take $K = \mathbb{R}^d$. Starting at x_0, repeat

$$\boxed{x_{t+1} = x_t - \eta \nabla f(x_t), \qquad t = 0, 1, \ldots.}$$

The parameter $\eta > 0$ is the step size.

4.1 Why move in the negative-gradient direction?

For a small step h,

$$f(x + h) = f(x) + \langle \nabla f(x), h \rangle + o(\|h\|_2).$$

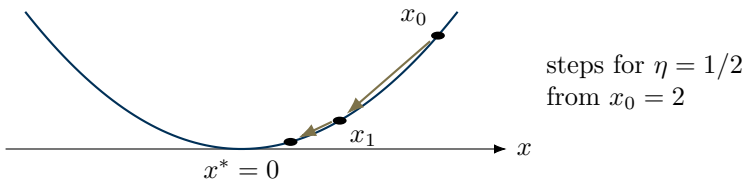
Among vectors with $\|h\|_2 \le \rho$, Cauchy–Schwarz gives

$$\langle \nabla f(x), h \rangle \ge -\rho \|\nabla f(x)\|_2.$$

If the gradient is nonzero, equality is attained in direction $-\nabla f(x)$. This is the steepest first-order decrease in Euclidean distance. The direction is useful, but the step length still matters.

Example. For $f(x) = x^2/2$, the update is $x_{t+1} = (1 - \eta)x_t$.

Step size	Behavior from $x_0 \neq 0$	Lesson
$\eta = 1/2$	$x_t = 2^{-t} x_0$	Converges
$\eta = 1$	$x_1 = 0$	Reaches the optimum immediately
$\eta = 2$	$x_t = (-1)^t x_0$	Oscillates without improving
$\eta > 2$	$\lvert x_t \rvert$ grows	Can diverge



A convergence theorem must specify assumptions, step sizes, *and which iterate or average is returned.*

5 The basic convergence guarantee

Assume $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable and convex, and has a minimizer x^*. Suppose

$$\|x_0 - x^*\|_2 \le D, \qquad \|\nabla f(x_t)\|_2 \le G \quad \text{at every iterate used.}$$

Here $D, G > 0$ are known bounds used to choose the step size. Fix an iteration count T, use the same η for all steps, and return the average of the points at which gradients were queried:

$$\bar{x}_T = \frac{1}{T}\sum_{t=0}^{T-1} x_t.$$

Theorem (gradient descent with bounded gradients). For every constant step size $\eta > 0$ satisfying the assumptions above,

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}.$$

Choosing $\eta = D/(G\sqrt{T})$ gives

$$\boxed{f(\bar{x}_T) - f(x^*) \le \frac{DG}{\sqrt{T}}.}$$

Thus $T \ge (DG/\varepsilon)^2$ suffices for additive error ε.

5.1 Proof plan before the algebra

We cannot evaluate x^*, but we can use it in the analysis. Track the squared distance

$$R_t = \|x_t - x^*\|_2^2.$$

The update expands R_{t+1} into R_t, a gradient-squared term, and an inner product. Convexity turns that inner product into $f(x_t) - f(x^*)$.

Our target one-step inequality is

$$\boxed{f(x_t) - f(x^*) \le \frac{R_t - R_{t+1}}{2\eta} + \frac{\eta G^2}{2}.}$$

Summing makes the distance terms telescope. Convexity then turns a bound on the *average function value* into a bound at the *average point*:

$$f\left(\frac{1}{T}\sum_{t=0}^{T-1} x_t\right) \le \frac{1}{T}\sum_{t=0}^{T-1} f(x_t).$$

This is the many-point version of the chord inequality.

5.2 Step 1: expand one update

Write $g_t = \nabla f(x_t)$. Then

$$\begin{aligned} R_{t+1} &= \|x_t - \eta g_t - x^*\|_2^2 \\ &= R_t - 2\eta\langle g_t, x_t - x^*\rangle + \eta^2\|g_t\|_2^2 \\ &\le R_t - 2\eta\big(f(x_t) - f(x^*)\big) + \eta^2 G^2. \end{aligned}$$

The last line uses precisely the first-order convexity inequality and the gradient bound. Rearranging proves the target one-step inequality.

5.3 Step 2: telescope

Sum over $t = 0, \dots, T-1$:

$$\begin{aligned} \sum_{t=0}^{T-1}\big(f(x_t) - f(x^*)\big) &\le \frac{1}{2\eta}\underbrace{\sum_{t=0}^{T-1}(R_t - R_{t+1})}_{R_0 - R_T} + \frac{\eta G^2 T}{2} \\ &\le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}, \end{aligned}$$

because $R_0 \le D^2$ and $R_T \ge 0$. Dividing by T and applying convexity to $\bar{x}_T$ proves the theorem.

5.4 Step 3: balance the two terms

For $\eta = D/(G\sqrt{T})$,

$$\frac{D^2}{2\eta T} = \frac{DG}{2\sqrt{T}}, \qquad \frac{\eta G^2}{2} = \frac{DG}{2\sqrt{T}}.$$

A tiny step keeps the discretization term small but makes progress slow. A large step does the opposite. The chosen step size balances the two.

A second valid output rule. If function values are available, choose $\hat{t} \in \arg\min_{0 \le t < T} f(x_t)$ and return $\hat{x} = x_{\hat{t}}$. This best queried iterate satisfies the same bound, since its value is at most the average value. This argument does *not* establish the same bound for the last iterate.

6 Scope, caveats, and what comes next

This is an additive approximation. The guarantee is $f(x) - f(x^*) \le \varepsilon$, not a multiplicative approximation ratio. The latter can be inappropriate when $f(x^*) = 0$ or is negative.

Count the work per iteration. The theorem bounds gradient-oracle calls. Total time also includes the cost of a gradient evaluation and vector arithmetic. For dense least squares with $A \in \mathbb{R}^{m \times d}$, computing $A^\top(Ax - b)$ takes $\mathcal{O}(md)$ time.

The bounds are real assumptions. Convexity alone does not bound D or G. For example, the gradient of $x^2/2$ is not bounded on all of $\mathbb{R}$. It is enough to bound gradients at the iterates, but the bound must be justified for the problem and step-size choice.

A short extension to nonsmooth functions. A subgradient g at x satisfies

$$f(y) \ge f(x) + \langle g, y - x \rangle \quad \text{for every } y.$$

Replacing gradients by subgradients with $\|g_t\|_2 \le G$ leaves the proof unchanged. For $f(x) = |x|$, use $g = 1$ for $x > 0$, $g = -1$ for $x < 0$, and any $g \in [-1, 1]$ at 0. This is a brief extension, not a new convergence proof.

Why projection and smoothness are next.

- A gradient step may leave a constrained feasible set K. We will project it back and preserve the distance analysis.
- A bounded gradient gives a $1/\sqrt{T}$ error bound. An upper bound on how rapidly the gradient changes will yield a faster $1/T$ guarantee.
- Lower as well as upper curvature bounds will give geometric convergence.

Exercise. With $D = 2$, $G = 3$, and $T = 900$, find the prescribed step size and the guaranteed additive error.

Exercise. Show that the average of T points in a convex set lies in that set. Then prove the many-point convexity inequality used in the analysis.

Exercise. For $f(x) = x^2/2$, check the potential calculation directly when $\eta = 1/2$. Which inequality in the general proof loses information specific to this quadratic?

CS 6515: Graduate Algorithms

Lecture 13: Projected and Smooth Gradient Descent

Instructors: Sahil Singla and Gerandy Brito

Fall 2026

Recap and plan. For convex f, bounded gradients, and $\|x_0 - x^*\|_2 \le D$, Lecture 12 proved

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}.$$

With $\eta = D/(G\sqrt{T})$, this is $DG/\sqrt{T}$. Today: keep iterates feasible using projection, then exploit upper curvature bounds to obtain faster convergence for a fixed objective.

1 Constrained optimization: project the step back

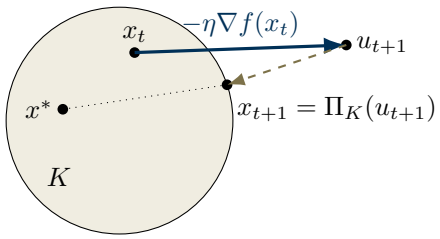
Let $K \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. The Euclidean projection of u onto K is

$$\Pi_K(u) = \arg\min_{z \in K} \|z - u\|_2^2.$$

The minimum exists, and its point is unique by strict convexity of squared distance on a convex set.

Projected gradient descent. Start at $x_0 \in K$, and repeat

$$u_{t+1} = x_t - \eta \nabla f(x_t), \qquad \boxed{x_{t+1} = \Pi_K(u_{t+1}).}$$



The raw gradient step can leave K; projection restores feasibility.

Example. For a box $K = \prod_i [a_i, b_i]$, project by clipping each coordinate:

$$(\Pi_K(u))_i = \min\{b_i, \max\{a_i, u_i\}\}.$$

Projection may itself be a nontrivial optimization problem. Its computational cost must be included in the running time.

1.1 Why projection preserves the distance argument

Theorem (projection cannot increase distance to a feasible point). If $p = \Pi_K(u)$ and $z \in K$, then

$$\boxed{\|p - z\|_2^2 \le \|u - z\|_2^2 - \|u - p\|_2^2 \le \|u - z\|_2^2.}$$

Proof. The segment $p + \theta(z - p)$ stays in K. Because p minimizes distance to u, the right derivative at $\theta = 0$ of $\|u - p - \theta(z - p)\|_2^2$ is nonnegative. Hence

$$\langle u - p, z - p \rangle \le 0.$$

Expand

$$\|u - z\|_2^2 = \|u - p\|_2^2 + \|p - z\|_2^2 + 2\langle u - p, p - z \rangle \ge \|u - p\|_2^2 + \|p - z\|_2^2.$$

Plug it into the existing proof. Take $z = x^* \in K$. Then

$$\|x_{t+1} - x^*\|_2^2 \le \|x_t - \eta \nabla f(x_t) - x^*\|_2^2.$$

Everything after this line is exactly the expansion and telescoping from Lecture 12. Therefore, with $\|\nabla f(x_t)\|_2 \le G$,

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}, \qquad \bar{x}_T \in K.$$

The same proof permits bounded subgradients.

2 Smoothness: an upper bound on curvature

For the faster bounds below, we return to **unconstrained** minimization over $\mathbb{R}^d$, with an attained minimum x^*.

Definition (smoothness). A differentiable $f : \mathbb{R}^d \to \mathbb{R}$ is β-smooth if

$$\|\nabla f(y) - \nabla f(x)\|_2 \le \beta \|y - x\|_2 \qquad \text{for all } x, y.$$

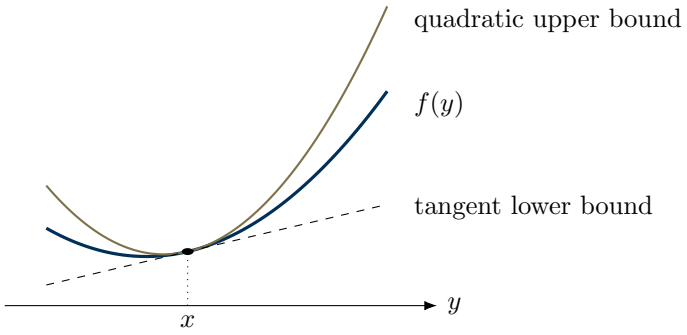
We use a positive upper bound $\beta > 0$.

Theorem (quadratic upper bound, or descent lemma). For a β-smooth function,

$$\boxed{f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2}\|y - x\|_2^2.}$$

Proof. Put $h = y - x$. The fundamental theorem of calculus and Cauchy–Schwarz give

$$\begin{aligned} f(y) - f(x) - \langle \nabla f(x), h \rangle &= \int_0^1 \langle \nabla f(x + sh) - \nabla f(x), h \rangle \, ds \\ &\le \int_0^1 \beta s \|h\|_2^2 \, ds = \frac{\beta}{2}\|h\|_2^2. \end{aligned}$$



For convex twice continuously differentiable functions, smoothness is equivalent to $0 \preceq \nabla^2 f(x) \preceq \beta I$. Smoothness bounds changes in the gradient, *not* its magnitude: $f(x) = x^2/2$ is 1-smooth on $\mathbb{R}$, but has unbounded gradient.

3 A safe step size and guaranteed descent

Choose $\eta = 1/\beta$:

$$x_{t+1} = x_t - \frac{1}{\beta}\nabla f(x_t).$$

Substituting this update into the quadratic upper bound gives

$$\begin{aligned} f(x_{t+1}) &\le f(x_t) - \frac{1}{\beta}\|\nabla f(x_t)\|_2^2 + \frac{\beta}{2}\frac{1}{\beta^2}\|\nabla f(x_t)\|_2^2 \\ &= f(x_t) - \frac{1}{2\beta}\|\nabla f(x_t)\|_2^2. \end{aligned}$$

Thus

$$\boxed{f(x_{t+1}) \le f(x_t) - \frac{1}{2\beta}\|\nabla f(x_t)\|_2^2.} \tag{1}$$

In particular, function values never increase.

Why the general-convex analysis was weaker. With only a gradient bound G, a fixed step need not decrease the function at every iteration. Smoothness controls the error in the linear approximation, allowing a constant safe step size and a last-iterate guarantee.

4 Smooth convex functions: a faster last-iterate bound

Theorem (smooth gradient descent). Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is convex and β-smooth, attains its minimum at x^*, and $\|x_0 - x^*\|_2 \le D$. After $T \ge 1$ steps of size $1/\beta$,

$$\boxed{f(x_T) - f(x^*) \le \frac{\beta D^2}{2T}.}$$

Thus $\mathcal{O}(\beta D^2/\varepsilon)$ iterations suffice for additive error ε.

Proof plan. Use the same squared-distance potential as before. This time the negative gradient-squared term from actual descent cancels the positive one in the distance expansion.

Set $g_t = \nabla f(x_t)$ and $R_t = \|x_t - x^*\|_2^2$. Expanding the update gives the exact identity

$$\frac{\beta}{2}(R_t - R_{t+1}) = \langle g_t, x_t - x^* \rangle - \frac{1}{2\beta}\|g_t\|_2^2.$$

By convexity and (1),

$$\begin{aligned} f(x_{t+1}) - f(x^*) &\le f(x_t) - f(x^*) - \frac{1}{2\beta}\|g_t\|_2^2 \\ &\le \langle g_t, x_t - x^* \rangle - \frac{1}{2\beta}\|g_t\|_2^2 \\ &= \frac{\beta}{2}(R_t - R_{t+1}). \end{aligned}$$

Sum for $t = 0, \dots, T-1$:

$$\sum_{t=0}^{T-1} \big(f(x_{t+1}) - f(x^*)\big) \le \frac{\beta}{2}(R_0 - R_T) \le \frac{\beta D^2}{2}.$$

Function values decrease, so each term on the left is at least $f(x_T) - f(x^*)$. Hence

$$T\big(f(x_T) - f(x^*)\big) \le \frac{\beta D^2}{2},$$

which proves the theorem.

Two improvements over the basic bound. The error is $\mathcal{O}(1/T)$, not $\mathcal{O}(1/\sqrt{T})$, and the guarantee holds for the last iterate x_T, not just an average. No extra $\log T$ factor is needed.

Assumptions	Output and error after T steps	Step size
Convex, gradients bounded by G	Average; $DG/\sqrt{T}$	$D/(G\sqrt{T})$
Convex and β-smooth	Last iterate; $\beta D^2/(2T)$	$1/\beta$

Do not overinterpret the conclusions.

- Convexity alone does not specify an efficient oracle or cheap projection. Running time includes those costs.
- For a nonconvex function, a zero gradient need not be a local minimum. Our global-optimality proofs used convexity essentially.
- A fixed-objective additive-error guarantee is not the same as an approximation ratio for a discrete optimization problem.

CS 6515: Graduate Algorithms

Lecture 14: Approximation Algorithms and Greedy Methods

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Fall 2026

Today's question. When exact optimization is computationally intractable, can we efficiently find a solution with a provable guarantee? We will use Vertex Cover as our central example and then study a short-path greedy algorithm for directed edge-disjoint paths.

1 Approximation guarantees

For an instance I, let $\mathrm{OPT}(I)$ denote the optimal value and (I) the value returned by an algorithm. The algorithm must return a *feasible* solution.

Definition (F). or $\alpha \geq 1$, an α-approximation satisfies

$$(I) \leq \alpha\,\mathrm{OPT}(I) \quad \text{(minimization)}, \qquad (I) \geq \frac{\mathrm{OPT}(I)}{\alpha} \quad \text{(maximization)}.$$

The guarantee holds for every instance, not just on average.

Approximation schemes. A PTAS accepts $\varepsilon > 0$ and produces a $(1+\varepsilon)$-approximation for minimization in time polynomial in the input size for each fixed ε. An FPTAS runs in time polynomial in both the input size and $1/\varepsilon$.

For maximization, the common convention is a guarantee of at least $(1-\varepsilon)\,\mathrm{OPT}$ (for $0 < \varepsilon < 1$). Recall the Knapsack FPTAS from earlier in the course.

Algorithmic theme. We will prove bounds by comparing our solution with a *certificate* or a structural bound on the unknown optimum.

2 Vertex Cover: two greedy algorithms

Given an undirected graph $G = (V, E)$, find a minimum-size set $C \subseteq V$ such that every edge has an endpoint in C. Write $n = |V|$ and $m = |E|$.

2.1 First greedy idea: choose the largest remaining degree

At each step, choose a vertex incident to the largest number of *uncovered* edges. Add it to the cover and remove all edges incident to it. Degrees are always measured in the graph of remaining edges.

Greedy-Degree-Cover(G)
$C \leftarrow \varnothing$; $E' \leftarrow E$
while $E' \neq \varnothing$:
  choose a vertex v incident to the maximum number of edges in E'
  $C \leftarrow C \cup \{v\}$
  remove from E' every edge incident to v
return C

Theorem (log approximation). For $m \geq 1$, Greedy-Degree-Cover returns a vertex cover of size at most $\mathrm{OPT}(1 + \ln m)$. Consequently it is an $O(\log n)$-approximation.

Proof. Feasibility is immediate: the algorithm stops only when every edge is covered. Fix an optimal vertex cover C^* with $|C^*| = \mathrm{OPT}$. Suppose $r > 0$ edges remain. The vertices of C^* still cover all r remaining edges. Counting incidences between these edges and vertices in C^*, some vertex of C^* is incident to at least r/OPT remaining edges. The greedy choice covers at least as many. Hence, writing r_t for the number of remaining edges after t choices,

$$r_{t+1} \leq r_t \left(1 - \frac{1}{\mathrm{OPT}}\right), \qquad r_t \leq m \left(1 - \frac{1}{\mathrm{OPT}}\right)^t \leq m e^{-t/\mathrm{OPT}}.$$

For $t > \mathrm{OPT} \ln m$, the right-hand side is less than 1, so the integer r_t must be zero. Thus the algorithm makes at most $\lfloor \mathrm{OPT} \ln m \rfloor + 1 \leq \mathrm{OPT}(1 + \ln m)$ choices. Since $m \leq n(n-1)/2 < n^2$ for a simple graph, $1 + \ln m = O(\log n)$. □

Interpretation. The number of uncovered edges decreases by a constant *fraction* in each step, but that fraction depends on $1/\mathrm{OPT}$. This is the familiar logarithmic greedy analysis from Set Cover, with edges as elements and vertices as sets.

2.2 A different greedy choice: build a maximal matching

The degree-based algorithm is very natural: we *cover* as many edges as possible in each step. Can we do better? One simple variation is to include two vertices at once by picking the edge with the most neighbors. We see next that even a relaxed version of this idea yields a better approximation.

A different greedy choice. Select an arbitrary uncovered edge (u, v) and take *both* endpoints. Delete all edges incident to either endpoint and repeat.

```
Greedy-Vertex-Cover(G)
C ← ∅; E' ← E
while E' ≠ ∅:
   choose any (u, v) ∈ E'
   C ← C ∪ {u, v}
   remove from E' every edge incident to u or v
return C
```

Theorem (2−approximation). Greedy-Vertex-Cover is a polynomial-time 2-approximation.

Proof. Let M be the edges chosen by the algorithm. No two edges of M share an endpoint, so M is a matching. At termination, every edge is incident to an endpoint of an edge of M, so C is a vertex cover. Every vertex cover must include at least one endpoint of *each* matching edge, and those edges are disjoint. Hence

$$|M| \leq \mathrm{OPT} \qquad \text{and} \qquad |C| = 2|M| \leq 2\,\mathrm{OPT}\,. \qquad \square$$

Terminology. The matching M is *maximal*: no further edge can be added to it. It need not be a *maximum* matching. Maximality suffices for feasibility; disjointness supplies the lower bound.

Tight example for this algorithm. On a single edge, the algorithm returns both endpoints, whereas an optimal cover contains one. Its factor 2 is attained.

Connection to duality. The matching M is a combinatorial lower-bound certificate: $|M| \leq \mathrm{OPT}$. Next lecture, an LP relaxation will provide another lower bound.

3 Directed edge-disjoint paths: greedy plus counting

Problem. Given a directed graph $G = (V, E)$ and terminal pairs (s_i, t_i) for $i = 1, \ldots, k$, route as many distinct pairs as possible. Choose at most one directed s_i–t_i path per pair, and require that every edge is used by at most one chosen path. Let $m = |E|$.

Unlike the single-source/single-sink version, this multicommodity optimization problem is computationally difficult. We seek a guarantee on the *number of routed pairs.*

Algorithm. Let $L = \max\{1, \lfloor\sqrt{m}\rfloor\}$ for $m \geq 1$. Repeatedly find any currently routable terminal pair with a directed path of length at most L (for example, use breadth-first search for shortest paths). Select such a path, remove its edges, and remove that pair from consideration. Stop when no such pair remains. Let A be the resulting set of paths.

Theorem (Disjoint Path Approximation). For $m \geq 1$, this algorithm returns at least $\mathrm{OPT}/(2\sqrt{m}+1)$ paths. In particular, it is an $O(\sqrt{m})$-approximation.

Proof plan. Fix an optimal collection O of edge-disjoint paths. Partition it into short paths (at most L edges) and long paths (more than L edges).

Long paths. Each long path uses at least $L+1$ edges. As the paths in O are edge-disjoint,

$$|O_{\text{long}}| \leq \frac{m}{L+1}.$$

Short paths. Consider a short optimal path Q for pair i. If greedy routes pair i, charge Q to the greedy path for that pair. Otherwise, because greedy has terminated, Q must share an edge with some selected greedy path; charge Q to one such path. The optimal paths are edge-disjoint, so at most L short optimal paths can be charged to a greedy path through shared edges. In addition, at most one path can be charged to it because it serves the same pair. Thus

$$|O_{\text{short}}| \leq (L+1)|A|.$$

If $\mathrm{OPT} > 0$, at least one terminal pair has a path of length at most L *or* all optimal paths are long. In the latter case A could be empty, so the short-path-only algorithm by itself does *not* give a multiplicative approximation. We fix this by also computing one arbitrary feasible path for one terminal pair, if any exists, and returning the larger of that one-path solution and A. Write $a \geq 1$ for the number of paths returned when $\mathrm{OPT} > 0$. Then

$$\mathrm{OPT} \leq (L+1)|A| + \frac{m}{L+1} \leq \left(L + 1 + \frac{m}{L+1}\right) a \leq (2\sqrt{m}+1)a.$$

(The last inequality uses $L = \lfloor\sqrt{m}\rfloor$ and $L + 1 > \sqrt{m}$, with $L + 1 \leq \sqrt{m} + 1$.) If $\mathrm{OPT} = 0$, the guarantee is trivial. □

Why the square root? The two losses compete: short-path charging costs about L per greedy path; the number of long optimum paths is at most m/L. Balance L and m/L by choosing $L \approx \sqrt{m}$.

Remark. The simple algorithm that repeatedly chooses the globally shortest available path also admits an $O(\sqrt{m})$ analysis. The cutoff version makes the short/long split explicit. We are *not* proving a hardness-of-approximation theorem today.

4 What to remember

1. A good approximation proof compares an efficiently constructed solution with a lower bound on OPT.

2. For Vertex Cover, a maximal matching certifies OPT $\geq |M|$ and gives a factor 2.
3. For edge-disjoint paths, a charging argument handles short optimal paths, and a counting argument handles long ones.
4. A bound tight for a particular algorithm is not automatically a hardness bound for all polynomial-time algorithms.

Next: Bridge to LP rounding. What if the lower-bound certificate is the optimum of a *linear relaxation*, rather than a matching or a counting argument?

CS 6515: Graduate Algorithms

Lecture 15: LP Relaxations and Deterministic Rounding

Instructors: Sahil Singla and Gerandy Brito

Fall 2026

Today's question. How can an optimal *fractional* solution help us find a good *integral* solution? We will first round Vertex Cover, then develop a 6-approximation for metric Facility Location.

1 The LP-rounding template

1. Formulate the discrete optimization problem as an integer program (IP).
2. Relax the integrality constraints to obtain a linear program (LP).
3. Solve the LP, obtaining an optimum x^* and a lower bound $^* \leq \mathrm{OPT}$ for minimization.
4. Round the fractional solution to a feasible integral one, bounding the cost increase.

The goal is to prove $\mathrm{cost}(\text{rounded solution}) \leq \alpha^* \leq \alpha\,\mathrm{OPT}$.

2 Vertex Cover: threshold rounding

Given $G = (V, E)$, introduce a variable x_v indicating whether vertex v is chosen. The integer program is

$$\begin{aligned} \text{minimize} \quad & \sum_{v \in V} x_v \\ \text{subject to} \quad & x_u + x_v \geq 1 \quad \forall \{u, v\} \in E, \\ & x_v \in \{0, 1\} \quad \forall v \in V. \end{aligned}$$

Relax to $0 \leq x_v \leq 1$. Let x^* be an optimal fractional solution, so $\sum_v x_v^* =^* \leq \mathrm{OPT}$.

Rounding rule. Return $C = \{v : x_v^* \geq 1/2\}$.

Theorem (Deterministic Rounding for Vertex Cover). Threshold rounding produces a vertex cover of size at most $2\,\mathrm{OPT}$.

Proof. For each edge $\{u, v\}$, the LP requires $x_u^* + x_v^* \geq 1$, so at least one endpoint has value at least $1/2$. Hence C covers every edge. Moreover, $\mathbf{1}[v \in C] \leq 2x_v^*$, and therefore

$$|C| = \sum_v \mathbf{1}[v \in C] \leq 2 \sum_v x_v^* = 2\,\mathrm{OPT}\,. \qquad \square$$

Connection to the matching algorithm. The LP dual (omitting redundant upper bounds) is

$$\begin{aligned} \text{maximize} \quad & \sum_{e \in E} y_e \\ \text{subject to} \quad & \sum_{e \ni v} y_e \leq 1 \quad \forall v \in V, \\ & y_e \geq 0 \quad \forall e \in E. \end{aligned}$$

A matching gives a feasible integral dual solution by setting $y_e = 1$ on matching edges and 0 elsewhere. Thus both lectures used lower-bound certificates, in two different forms.

Integrality gap. In the complete graph K_n, the integral optimum is $n-1$, while the fractional optimum is $n/2$ (set each $x_v = 1/2$; optimality also follows by summing the edge constraints). The ratio $2 - 2/n$ approaches 2. The LP itself cannot generally certify a better factor than 2.

3 Facility Location

Let F be candidate facilities and C clients. Opening facility $i \in F$ costs $f_i \geq 0$. Assigning client $j \in C$ to facility i costs $d_{ij} \geq 0$. Distances are induced by a metric on $F \cup C$, so the triangle inequality applies.

Choose facilities $S \subseteq F$ to open and connect each client to an open facility. Minimize

$$\sum_{i \in S} f_i + \sum_{j \in C} \min_{i \in S} d_{ij}.$$

Integer program. Let $x_i = 1$ mean facility i is open, and $y_{ij} = 1$ mean client j is assigned to i:

$$\begin{aligned} \text{minimize} \quad & \sum_{i \in F} f_i x_i + \sum_{i \in F, j \in C} d_{ij} y_{ij} \\ \text{subject to} \quad & \sum_{i \in F} y_{ij} = 1 \quad \forall j \in C, \\ & y_{ij} \leq x_i \quad \forall i \in F,\ j \in C, \\ & x_i, y_{ij} \in \{0, 1\}. \end{aligned}$$

Relax integrality to $0 \leq x_i, y_{ij} \leq 1$, and solve for (x^*, y^*). Define

$$F^* = \sum_i f_i x_i^*, \qquad D^* = \sum_{i,j} d_{ij} y_{ij}^*, \qquad OPT^* = F^* + D^* \leq \text{OPT}\,.$$

3.1 Deterministic Rounding for Facility Location

Let (x^*, y^*) be an optimal solution to the LP relaxation. Recall that x_i^* represents the fractional opening of facility i, and y_{ij}^* represents the fraction of client j assigned to facility i.

For each client $j \in C$, define

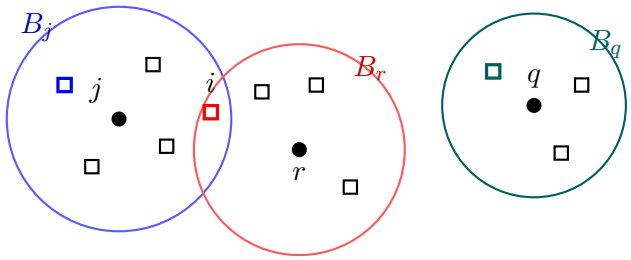
$$L_j = \sum_{i \in F} d_{ij} y_{ij}^*.$$

We interpret L_j as the average distance from client j to the facilities serving it in the fractional solution.

Define the ball

$$B_j = \{i \in F : d_{ij} \leq 2L_j\}.$$

Thus B_j contains all candidate facilities within distance $2L_j$ of client j.



The balls B_j and B_r overlap because they share a candidate facility i. The ball B_q is disjoint from both. Squares represent candidate facilities and dots represent clients.

3.2 The Rounding Algorithm

We process clients in nondecreasing order of L_j. The set R will contain the representatives of the balls selected by the algorithm.

Algorithm: Facility Location Rounding

1. Sort clients so that
$$L_{j_1} \leq L_{j_2} \leq \cdots \leq L_{j_{|C|}}.$$
2. Initialize $R = \emptyset$ and the set of open facilities $S = \emptyset$.
3. For each client j, in the above order:
   (a) If $B_j \cap B_r = \emptyset$ for every $r \in R$, choose the cheapest facility in B_j:
   $$i_j \in \arg\min_{i \in B_j} f_i.$$
   Open facility i_j, add i_j to S, add j to R, and assign j to i_j.
   (b) Otherwise, choose some $r \in R$ with $B_j \cap B_r \neq \emptyset$ and assign client j to the facility i_r opened for r.
4. Return the opened facilities and client assignments.

Notice that the balls corresponding to clients in R are pairwise disjoint. This property is essential for bounding the opening costs.

Next time we will prove that this algorithm yields a $6-$approximation.